



CS-317 OPERATING SYSTEMS

Lecture 3



PROCESSES

Book 1 Chapter 3

Things we learned in the last lecture...

- In the last lecture, we introduced the concept of a process. A process is a program in execution.
- We saw the reasons due to which a new process is created, and the reasons which can lead to its termination.
- We introduced the concept of virtual memory, and suspension.
- We traced the life of a process from birth to death and explored the different states that it can have in an OS.



PROCESS DESCRIPTION

Data structures that describe a process

The Multiprogrammed system

- There are n processes in a virtual memory system.
- Shared system resources include the processor, I/O devices and main memory.
- Take a moment to guess:
 - Which process is running?
 - Which process is suspended?

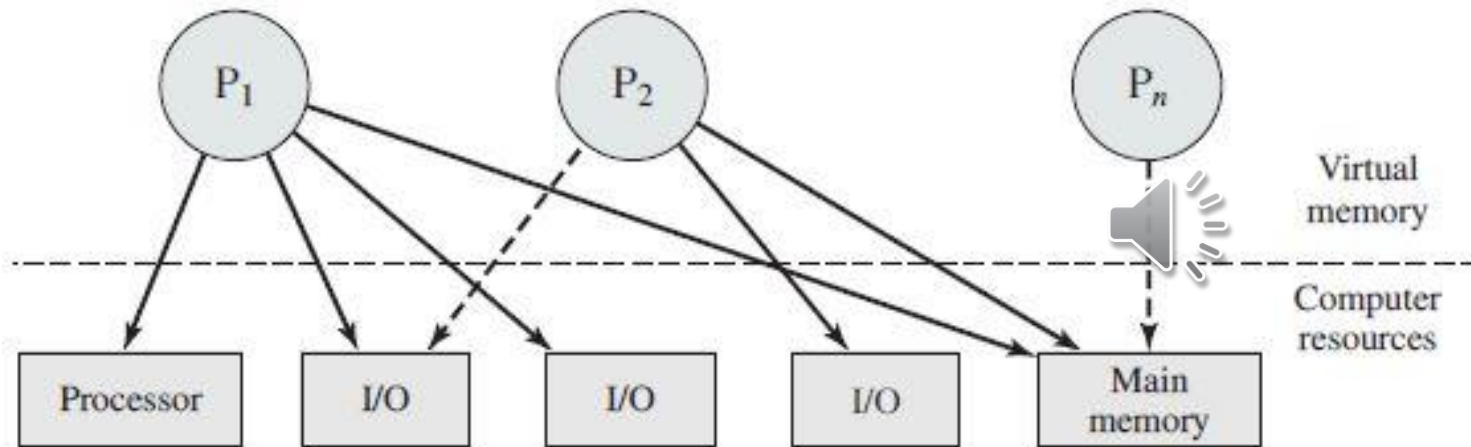
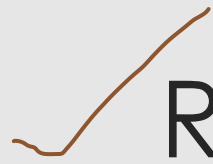


Figure 3.10 Processes and Resources (resource allocation at one snapshot in time)



→ it is directly connected to the processor, so it is currently executing.
→ it is only linked to main memory (virtual memory) with a dashed line, indicating it is not active (suspended).
P₂ is in the Blocked (Waiting) state.



Role of OS in a Multiprogrammed system

A multiprogrammed system is an operating system technique where multiple processes are kept in memory at the same time, and the CPU switches between them to maximize usage.

- Create and manage processes
- Schedule and dispatch processes to the processor
- Allocate and deallocate virtual and primary memory spaces to processes
- Allocate I/O resources to processes
- Respond to OS service calls by processes



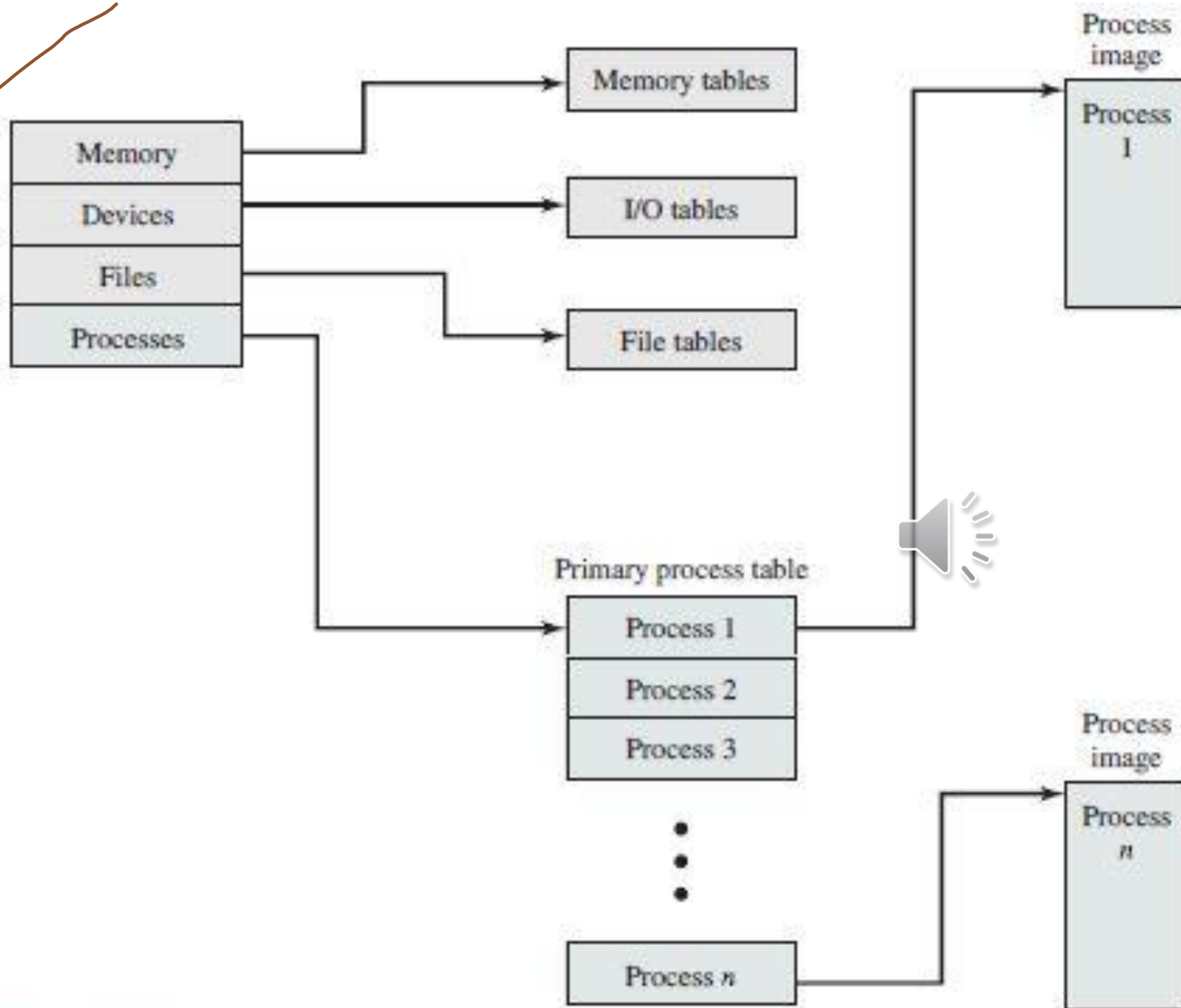


Figure 3.11 General Structure of Operating System Control Tables

OS control structures

- Memory tables
- I/O tables
- File tables
- Process tables

Memory tables

protection attributes are :
Which processes are allowed to read, write, or execute it
Whether the memory is shared or private
Any special restrictions (e.g., read-only, kernel-only)

this tells for each block

- These tables are used to keep track of both primary and secondary storage.
- A part of main memory is reserved for the OS, the **system address space**.
- The rest of main memory is used by the processes and is called the **user address space**.
- All processes are completely loaded in the hard disk portion of virtual memory.
- Memory tables must include these pieces of information:
 - Allocation of main memory to processes,
 - Allocation of secondary memory to processes,
 - Protection attributes of blocks of main or virtual memory, such as which processes may access shared memory regions,
 - Any information needed to manage virtual memory.

I/O tables

- These tables are used to manage I/O devices.
- I/O tables must include these pieces of information:
 - Availability or assignment of I/O devices,
 - Status of an I/O operation in progress,
 - Location of main memory used as source or destination of an I/O transfer



File tables

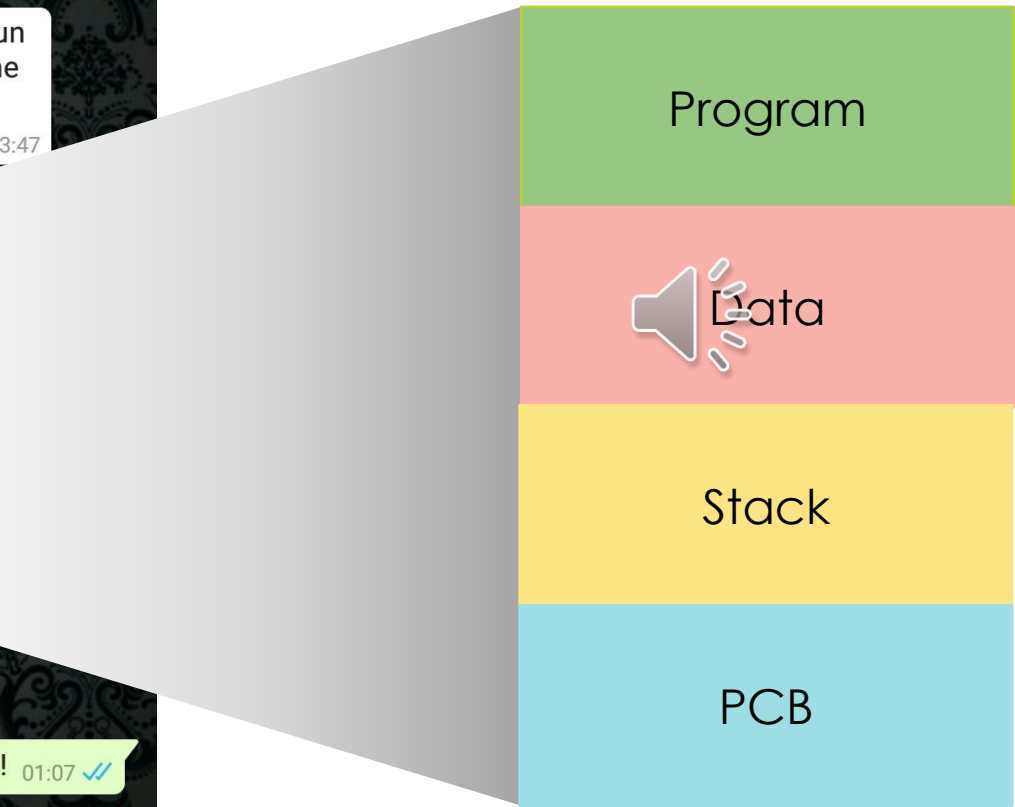
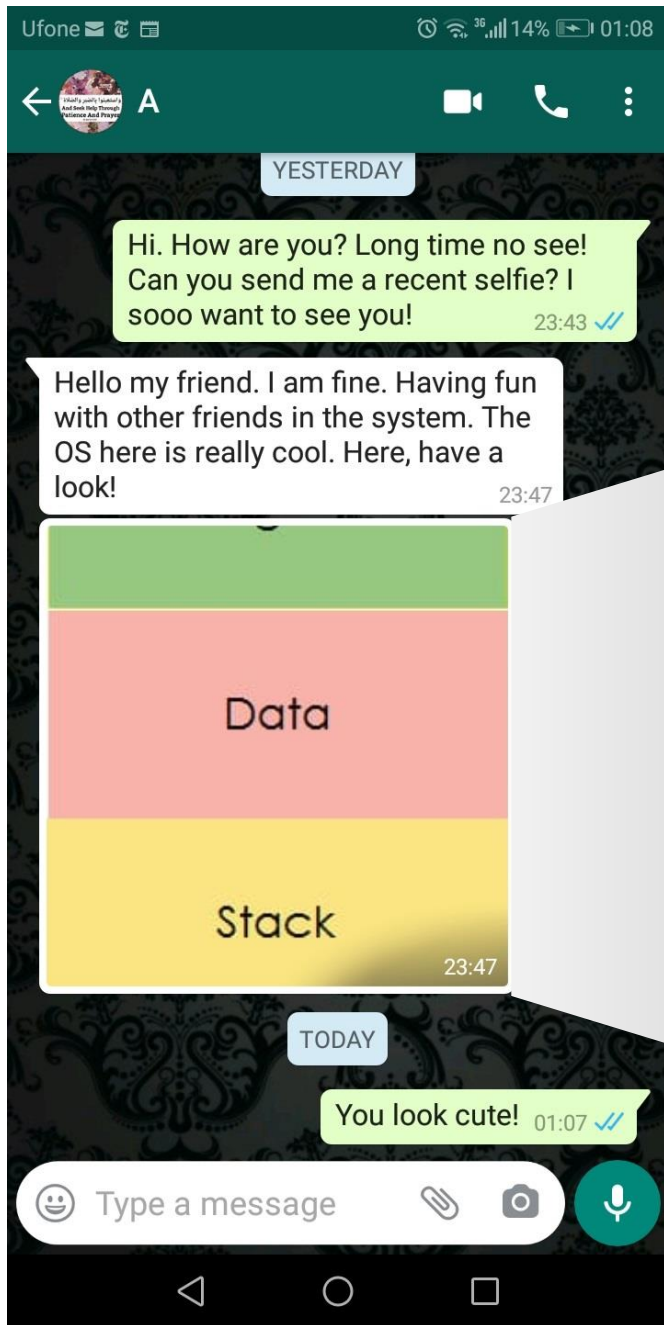
- These table are used to manage files in main and secondary storage.
- File tables must include these pieces of information:
 - Existence of files
 - Location of files on secondary storage
 - Status of files
 - File attributes shared access, read write access
- In some operating systems, there is a separate file management system, so the OS does not bother about files.

Process tables

- These table are used to manage processes that exist in the system.
- The topic of process description deals with process tables and data structures that help the OS to manage processes.
- These tables must include the following information:
 - Location of processes
 - Attributes of processes

Process Image

- The process image is a collection of things that are kept in the address space that the OS assigns when it creates a process.
- There are four things that are kept in the process' address space:
 - **User program...** the program to be executed
 - **User data...** values that the user has provided as individual variables or in complex data structures like arrays, matrices, linked lists, trees, etc.
 - **Stack...** to handle procedure and system calls
 - **Process Control Block...** the process' ID card
- If the process is suspended, all pages of the process image are on the hard disk. Otherwise, at least one page of the process image is loaded in the main memory.
- The OS needs to know the locations of all pages of the process image to be able to fetch and execute the process. This information is maintained by the memory management part of the OS, which keeps the memory tables.



Process Image

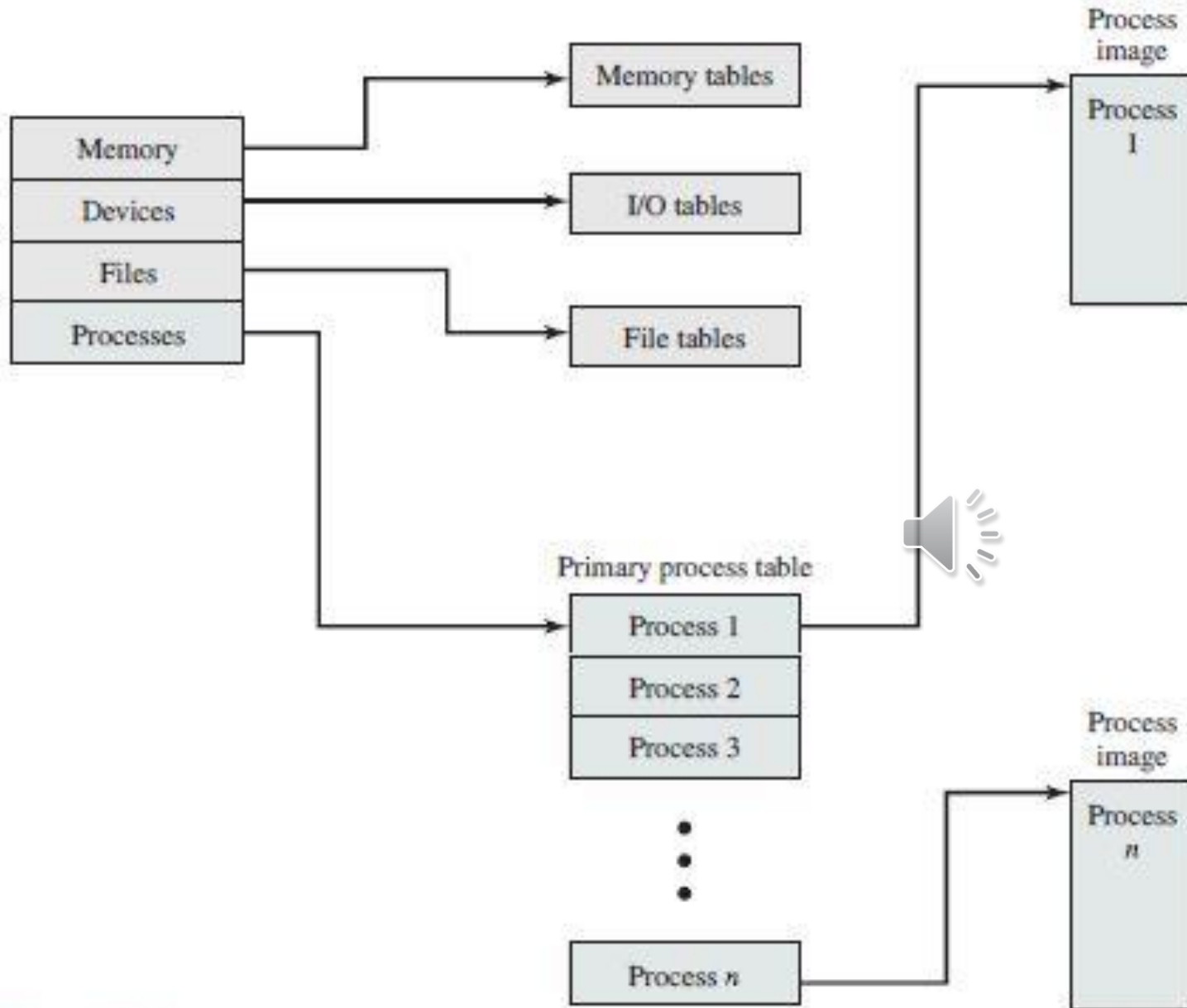


Figure 3.11 General Structure of Operating System Control Tables

OS control structures

- Memory tables
- I/O tables
- File tables
- Process tables

Process Control Block (PCB)

- The OS generates a PCB for each process that it creates. This data structure has a lot of sub fields.
- We can characterize information contained in a PCB in three broad categories:
 - Process identification
 - Processor state information
 - Process control information



Process identification



This section of the PCB stores various identifiers relevant to the process.



It contains information like:

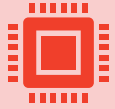
Process ID

User ID

Parent ID



Processor state information



This section of the PCB contains contents of the processor's registers.



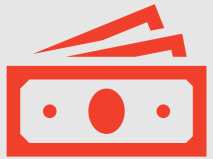
This is the **context** of the process which is **saved** when it is preempted from the processor and **restored** when the processor is allocated to it.



For any processor, the register set includes:

User visible registers
Control and status registers
Stack pointers

Control and status registers



Program Counter



Condition codes

Sign
Zero
Carry
Equal
Overflow



Status information

Interrupt flag
Execution mode

Task# 1: In the EFLAGS register, identify the bits relevant to interrupts and execution mode.

IF (bit 9): Controls interrupts (1 = enabled, 0 = disabled)
IOPL (bits 12–13): Defines privilege level for execution/I/O
VM (bit 17): Enables Virtual 8086 mode



Task# 2: In the EFLAGS register, identify the bits relevant to condition codes.

Pause the video!

Write your answer in a word file. Save it as <my_roll_no>_lecture_3. If your roll no is 500, the file will be named 500_lecture_3. Do not add <> signs to the filename.

CF (Carry Flag – bit 0): Indicates carry/borrow in arithmetic
PF (Parity Flag – bit 2): Set if result has even parity
AF (Auxiliary Carry – bit 4): Carry from lower nibble (BCD operations)
ZF (Zero Flag – bit 6): Set if result is zero
SF (Sign Flag – bit 7): Indicates sign of result (1 = negative)
OF (Overflow Flag – bit 11): Set if signed overflow occurs

Task# 3: Can you define parity?

Parity means whether the number of 1s in a binary value is even or odd.

Task# 4: Can you define overflow?

Overflow occurs when the result of a signed arithmetic operation is too large or too small to fit in the given number of bits.

Pause the video!

Write your answer in a word file previously saved as <my_roll_no>_lecture_3.



Process control information



This section of the PCB contains information used to coordinate and control various active processes in the system.



It includes the following information:

- Scheduling and state information
- Data structuring
- Interprocess communication
- Process privileges
- Memory management
- Resource ownership and utilization



Scheduling and state information

This information is needed to properly schedule the process.

This section include information like:

- Process state,
- Priority,
- Event,
- Scheduling related information like:
 - Execution time of the process in its last run,
 - Waiting time of the process, etc.

Data structuring

A process may be linked to other processes in:

- A queue,
- A ring,
- A tree, etc.



This section contains pointers to other processes to support these structures.



Interprocess communication

Communication between processes in a system is called Interprocess communication.

- When a system has many processes, we can have a scenario where one process generates data, and another process or a set of processes use it. If this happens, the generating process will have to send data to the using processes. This will lead to data transfer or communication between processes in the same system.

Interprocess communication uses:

- Flags,
- Variables, or
- Messages.

Some or all of this information may be kept in this section of the PCB.

✓ Memory management



In a virtual memory system, this section includes pointers to:

- Segments or
- Pages, of the process.



Resource ownership and utilization

Anything that a process may ask for is a resource.

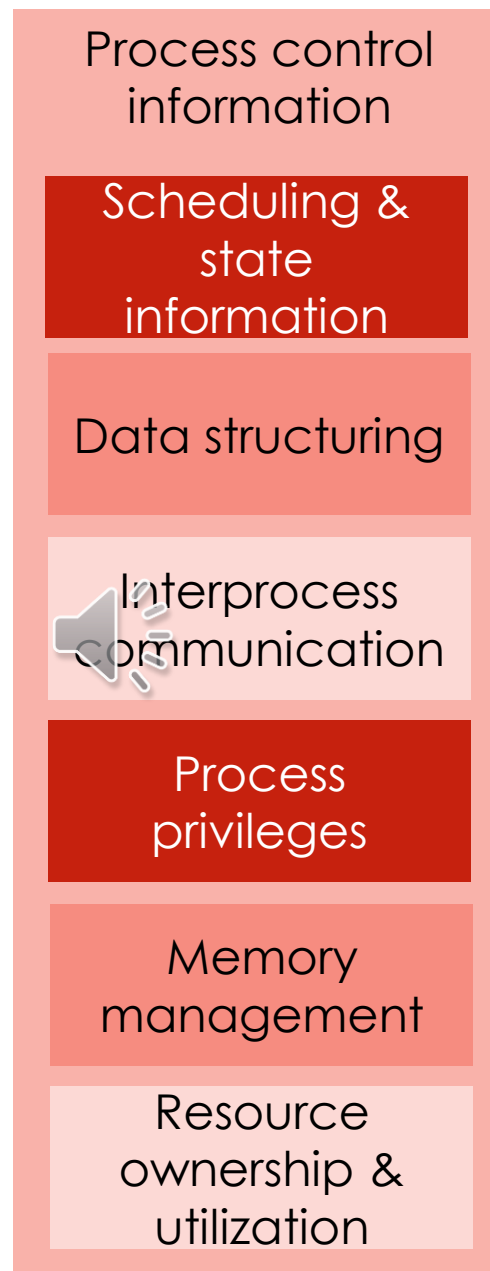
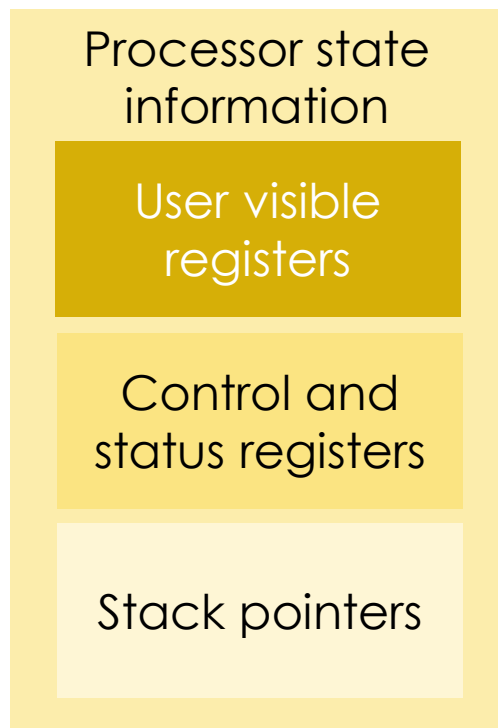
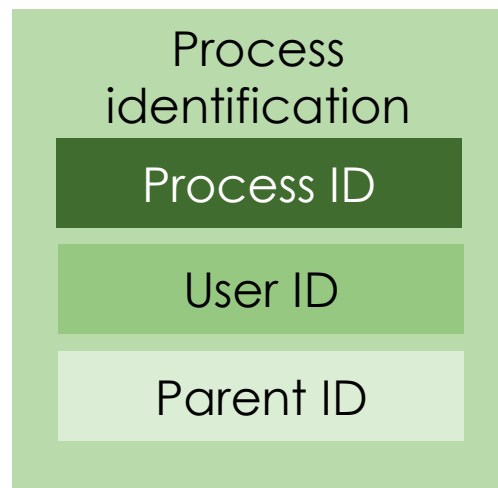
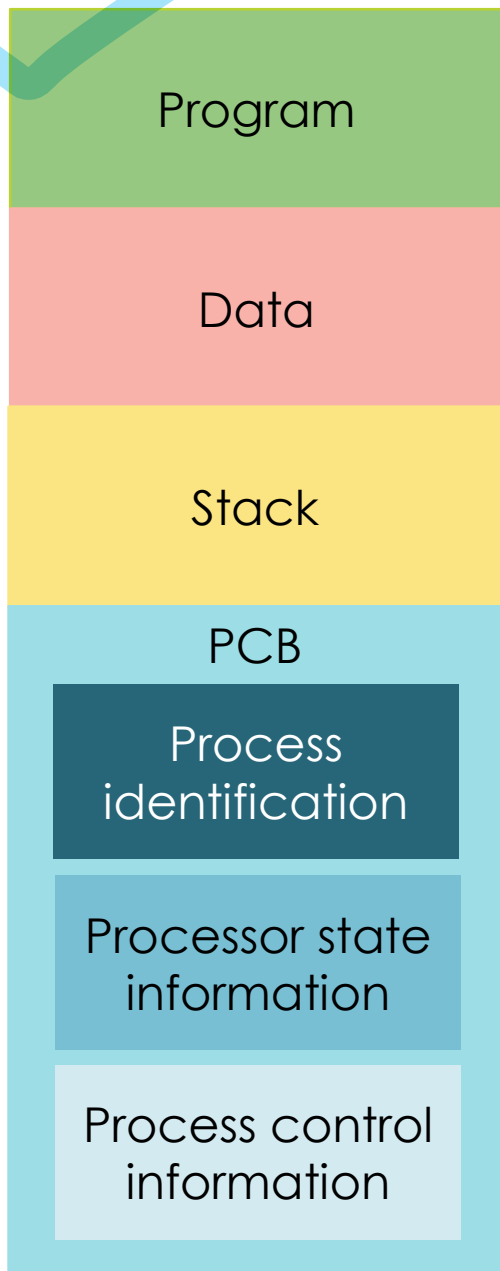
System resources include:

- Processor
- Memory
- Files
- Variables
- Messages
- I/O devices, etc.



This section contains IDs of the resources that have been granted to the process.

Resource utilization history may also be maintained.



Process description: The complete picture

I have drawn here for you:

- The process image
- The three parts of PCB
- Individual sections of PCB with their constituents

Process privilege (or process privilege level) means the level of access and permissions a process has to use system resources like memory, CPU, files, and devices.
high privilege = kernel mode
low = user mode

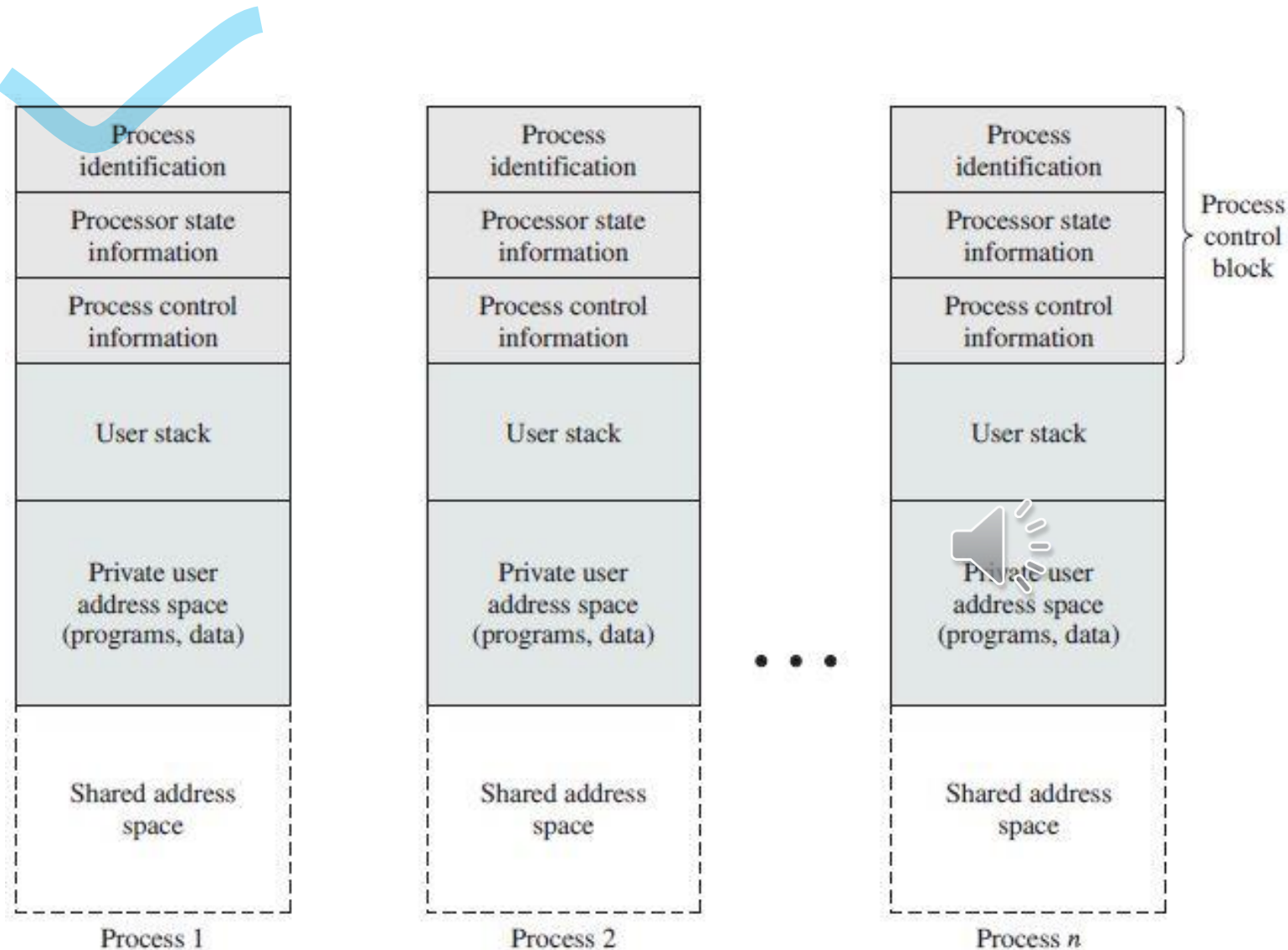


Figure 3.13 User Processes in Virtual Memory

User processes in virtual memory

- Here process images are contiguous, but it is not necessary that all pages of a process lie contiguously in virtual memory.
- There is an area of memory that a process may share with another process, and shared variables can be kept here.

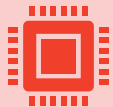
The role of PCB



It is the most important data structure in the OS.



The set of all PCBs in a system defines the state of the OS.



Every OS module reads and/or writes the PCB.

Problems with direct access of PCBs

If each OS module directly accesses all PCBs,

- A bug in a single OS routine can damage all PCBs in the system.
- A design change in the PCB can affect many OS modules.



Make a PCB handler routine.



All OS modules wanting to read or write PCBs must ask the PCB handler to do this on their behalf.



Only a bug in the PCB handler will affect PCBs.



If the design of the PCB is changed, only the PCB handler has to be changed.



The system performance suffers, as the PCB handler becomes a performance bottleneck.



Solution

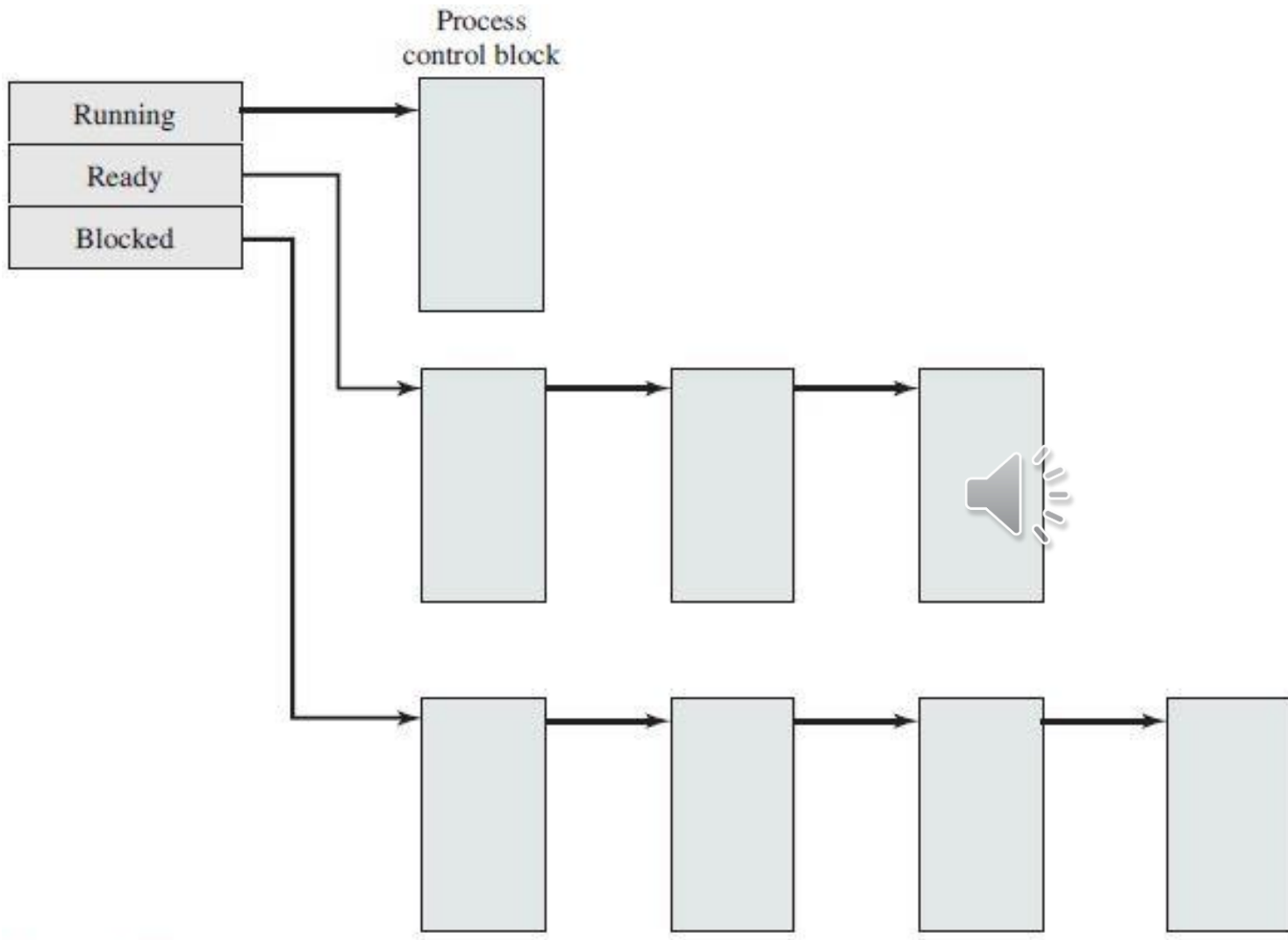


Figure 3.14 Process List Structures

Process list structures

- How many processes are ready?
- How many are blocked?
- Here processes with the same state are arranged using a data structure. Can you name it?

Number of ready processes → length of ready list
Number of blocked processes → length of blocked list

data structure = linkedlist

Concluding remarks

- In this lecture we understood the process image.
- We talked about the Process Control Block in detail.
- We understood the data structures the OS uses to manage processes.
- We realized the role of the PCB in an OS.

Things to do

- Read the book.
- I will ask you to submit the word file previously saved as <my_roll_no>_lecture_3 in  Google classroom.
- Write your questions on a piece of paper and keep it in front of you during the online session.
- If you have suggestions relevant to content or quality of this lecture, email them to me.